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Power System Stability with High Penetration of Renewable Energy Sources: Challenges, Assessment, and Mitigation Strategies

Jana Sheikh Ali¹, Yazan Qiblawey², (Member, IEEE), Abdulrahman Alassi², (Senior Member, IEEE), Ahmed M. Massoud¹, (Senior Member, IEEE), S M Muyeen¹, (Fellow, IEEE), Haitham Abu-Rub³, (Fellow, IEEE)

¹Department of Electrical Engineering, Qatar University, Doha, Qatar,

²Iberdrola Innovation Middle East, Doha, Qatar

³Department of Electrical and Computer Engineering, College of Science and Engineering, Hamad Bin Khalifa University, Qatar Foundation, Doha, Qatar

Corresponding author: Jana Sheikh Ali (e-mail: js1909504@student.qu.edu.qa).

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ABSTRACT The incorporation of renewable energy sources (RESs) into power systems has significantly increased in recent years due to growing environmental, economic, and energy security concerns, posing new challenges for power system stability. This paper first presents a modified stability categories classification for power systems with increased RES penetration, distinguishing between fundamental and non-fundamental stability. Moreover, the paper comprehensively surveys the existing techniques for evaluating power system stability (PSS) with high RES penetration. It highlights key factors such as the level of RES penetration, network topology, assessment and mitigation strategies, and the simulation tools used. The PSS methodologies in the literature are categorized into three main categories: model-based, optimization-based, and artificial intelligence (AI)-based approaches. This survey aims to equip researchers and practitioners with a coherent framework for navigating PSS assessment techniques. It offers promising avenues and insights into future research directions and related challenges, particularly addressing PSS with increased RESs penetration.

INDEX TERMS Power system stability, high renewable energy penetration, model approach, optimization, AI-based approach.

I. INTRODUCTION

The increasing integration of renewable energy sources (RESs) into power systems has raised concerns about preserving system stability due to their fluctuating and unpredictable nature [1]. This shift from fossil fuel-based generators to RESs – such as wind turbines and solar photovoltaics (PV) – is driven by the urgent need to reduce greenhouse gas emissions and combat climate change [2]. For instance, the European Parliament and Council's tentative agreement aims for a 45% RES penetration by 2030 [3]. While this transition is crucial for a sustainable future, it also presents new challenges for power system stability (PSS), emphasizing the need for effective strategies to ensure robust and reliable power grids amidst increasing RES integration.

A. CHALLENGES OF INCREASED RENEWABLES PENETRATION

Although increasing the use of RESs in the electrical grid has a positive environmental impact compared to traditional fossil fuel-based energy generation, it has created several problems, particularly those related to the grid's stability. Ref. [4] outlined several key challenges: firstly, the output of RESs fluctuates due to the natural variability of their primary source. Furthermore, RESs tend to be unpredictable, leading to high uncertainty within the system. Another issue is that RESs must be constrained by location depending on natural resources and infrastructure availability. One of the most critical issues is that, unlike traditional synchronous generators, RESs have a very low inherent inertia, which

weakens their ability to mitigate frequency deviations. This vulnerability makes the power system more susceptible to instability.

In addition to the general challenges presented by the integration of RES in the grid, the challenges can also greatly depend on geographical regions. In some regions, heavy reliance on a single energy source, such as hydropower, can be limiting due to its variable and seasonal nature, while geopolitical and economic barriers may limit the expansion of solar and wind energy [5]. Additionally outdated infrastructure and insufficient investment further impedes RES integration, particularly in developing regions where financing and grid modernization efforts are constrained [6]. The primary focus of this paper is to investigate the PSS in the context of increased RES penetration, emphasizing challenges, assessment, and mitigation strategies.

B. RELEVANT WORKS

Several review papers have addressed the challenges of PSS for power systems with RES penetration. Some review papers aimed to classify relevant literature into appropriate categories [7], [8]. For instance, Ref. [7] explored the growing PSS challenges, classification, and research opportunities in the context of a high proportion of RESs and power electronics. The paper summarised how the increasing use of RESs affects classical stability problems and identified new stability challenges. It also reviewed the existing classical and extended PSS classifications, highlighted their limitations, and proposed a new classification for PSS categories. Ref. [8] has extensively reviewed modeling principles and mathematical analysis methodologies for transient stability in large-scale power systems. The review divided transient stability analysis methodologies into Lyapunov-based and non-Lyapunov-based methods and comprehensively reviewed each technique.

Similarly, the primary focus of several other relevant review papers was to emphasize the challenges and effects of increasing RES penetration in power systems [7 - 10]. For instance, the technical difficulties that arise in power systems with minimal inertia when integrating up to 100% inverter-based resources (IBRs) have been thoroughly examined in Ref. [9]. The study provided an overview of the dynamic stability difficulties that affect grid operations and discussed the solutions employed in literature to minimize these challenges. Ref. [10] investigated the implications of increased PV penetration on the stability and dependability of power networks. It examined the effects of PV integration on various factors, including voltage, frequency, protection, harmonics, rotor angle stability, and flexibility requirements in power systems. The research also focused on the tools and models often used in power system analysis to manage the implications of high PV penetration. Ref. [11] provided a detailed review of grid frequency stability issues associated with the increasing integration of IBRs from the perspective of system operators. It investigated the effect of IBR on frequency stability and the management of Rate of Change of Frequency (RoCoF) in weak power networks. Stability concepts and challenges in the power electronics-dominated

grid (PEDG) paradigm were discussed in [12], highlighting the specific PEDG network characteristics, including low inertia and stochastic generation. The paper categorized the stability of PEDG into supply-demand and control stability, covering issues such as power fluctuations, voltage oscillations, and frequency variations. Moreover, open research questions about the transition from traditional power networks to PEDG were outlined, focusing on the role of inverters and different grid operation modes.

Other papers focused on reviewing the control approaches found in the literature to enhance PSS with increased RES penetration [11 - 14]. Ref. [15] reviewed the most recent oscillatory stability monitoring, analysis, and control approaches published in the literature, specifically focusing on synchrophasor technology. The study addressed oscillation challenges in power networks with significant RES penetration and presented an improved classification for power system oscillations, with a particular emphasis on those caused by IBRs. These control approaches, when implemented, can significantly enhance the stability and reliability of power systems with high RES penetration, making them a valuable resource for researchers and professionals in the field.

Machine learning (ML) techniques have also been applied for PSS [15 - 17]. Ref. [17] has reviewed recent research on applying ML techniques to improve power system security and stability. The study examined how these techniques detect cyberattacks, investigate power quality issues, and assess dynamic security. It also discussed the methodologies, successes, and constraints of designing classifiers, generating datasets, and testing systems utilized in these investigations. These practical applications of ML techniques in power system security and stability underscore their potential in addressing the challenges of increasing RES penetration, making them a valuable resource for researchers and professionals. Ref. [19] presented a comprehensive review focusing on the application of deep reinforcement learning (DRL) for PSS. The paper highlighted that the rising integration of RES is leading to uncertainties and complexities in modern grids, and how DRL can be used to provide near optimal control strategies for such issues. Moreover, a detailed analysis of the strengths, limitations, and practical challenges of different DRL approaches were discussed.

Based on the discussion above, it is evident that extensive research endeavors have been conducted regarding PSS in the context of RES penetration. However, few of these works provided a comprehensive review that focuses on categorizing the different methodologies used to study PSS with RES penetration, emphasizing different categories of PSS, challenges, assessment, and mitigation strategies. To bridge this research gap, this review paper intends to classify the various methods used for PSS assessment while highlighting the percentage of RES penetration, PSS types, simulation of the penetration level and impact, network topology, assessment and mitigation approaches, and applied simulation tools.

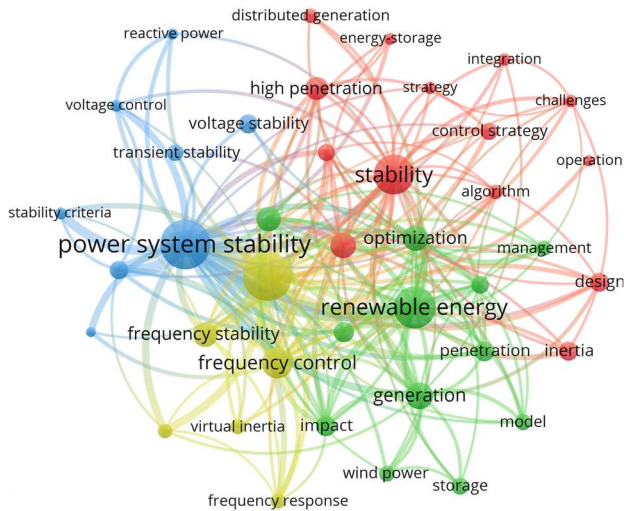


FIGURE 1. Keyword map for PSS analysis with high RES penetration.

C. PAPER CONTRIBUTIONS

Surveying and examining different approaches documented in the literature to improve PSS with RES penetration is essential. Using the Web of Science (WoS) core collection database and VOSviewer software, a bibliometric analysis was carried out to organize the relevant data on this topic [20]. The database was searched with main keywords related to the topic, including "power system stability" and "renewable energy penetration". Papers published within the last five years were prioritized, although older studies were considered based on relevance. Studies focusing on power electronic converter stability or those not addressing RES integration were excluded. The bibliometric analysis results are shown in Fig. 1. In the diagram, the circles represent nodes, with their size indicating how frequently a specific keyword was used in literature. The paper's primary contributions are highlighted as follows:

1. Present a modified stability issues classification for networks with increased RES penetration, highlighting the difference between fundamental and non-fundamental component stability.
2. Develop a systematic framework to classify and analyze the methods for PSS with high RES penetration, categorizing them into three main groups: model-based, optimization-based, and AI-based.
3. Conduct a comparative evaluation of the different techniques based on PSS categories (identifying the specific type of stability issue addressed by each method), RES penetration quantification (assessing the method's capability to quantify and model the impact of different levels of RES penetration), and network case studies analyzing the types of networks used in the evaluation of each technique.

The paper's structure is illustrated in Fig. 2. This paper is divided into the following sections: Section II establishes the fundamental types of PSS issues relevant to increasing RES penetration. Section III discusses the model-based approaches for PSS with increased penetration of RESs, while Section IV focuses on the optimization methods.

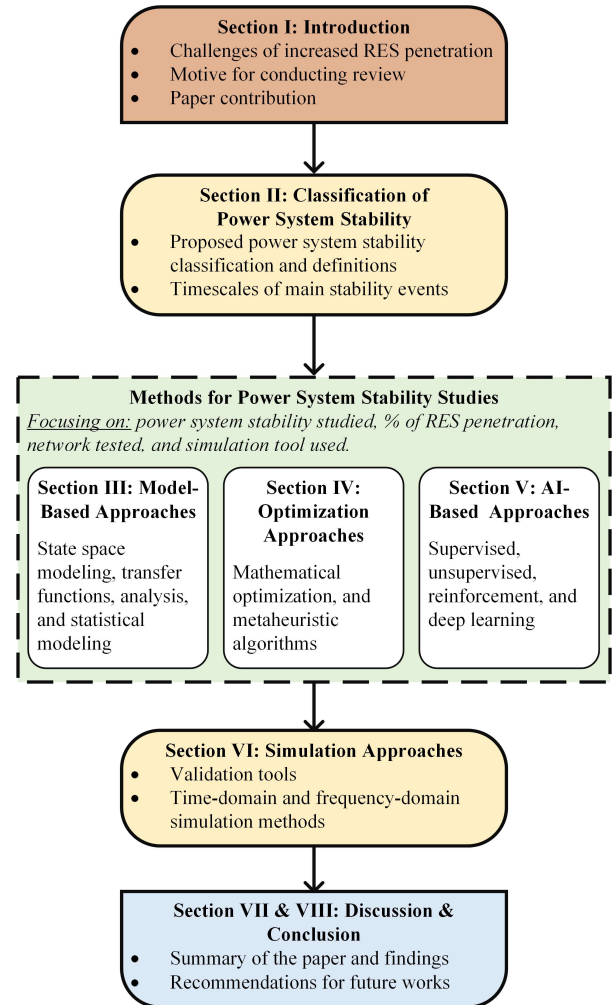


FIGURE 2. Structure of the paper.

Section V focuses on AI applications for PSS, Section VI presents a discussion of the findings, while Section VII offers a conclusion and summarizes the overall review.

II. CLASSIFICATION OF POWER SYSTEM STABILITY

Classical PSS is typically divided into three main categories: rotor angle stability, voltage stability, and frequency stability [21]. To study the effect of the increased RES penetration in the electrical grid, a modified stability classification is illustrated in Fig. 3. This classification, intended to align with the IEEE classification presented in [22], outlines the various stability issues that may arise in networks with increased RES penetration. The modified classification is divided based on the frequency range of the stability event: fundamental component stability addresses stability issues around the operating system frequency, while non-fundamental component stability includes stability events that occur at different frequency ranges. Non-fundamental component stability dynamics are often not considered within the scopes of traditional frameworks. This modified classification improves on current frameworks as it bridges the gap between traditional and modern power systems by taking into consideration the growth of RESs in

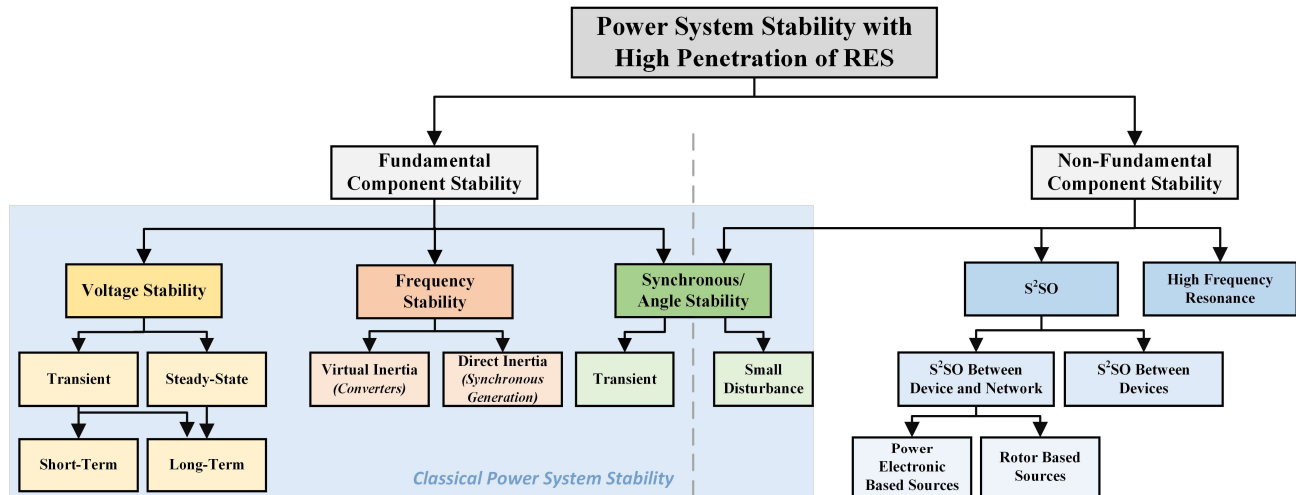


FIGURE 3. Modified classification of the categories of power system stability (S²SO denotes sub/super-synchronous stability).

contemporary power systems and the specific stability issues that they pose. The classical power system stability categories, highlighted in blue in Fig. 3, are the main focus of this paper, while non-fundamental component stability, being a more specific issue, will be addressed separately in future publications. To better illustrate the differences between the main stability categories, Fig. 4 was generated based on the information from the IEEE classification [22]. The displayed time frames correspond to the range of time over which the different types of dynamic behaviors happen, based on the power system dynamic phenomena.

A. SYNCHRONOUS/ANGLE STABILITY

The ability of synchronous machines in traditional power systems to retain synchronism following disruptions is referred to as rotor angle stability. It can be divided into two categories: transient stability, which deals with major disturbances altering rotor angles, and small-disturbance stability, which handles subtle changes like load shifts [23]. In RES-based grids, rotor angle stability is maintained using virtual synchronous machines, known as synchronverters, by emulating inertia.

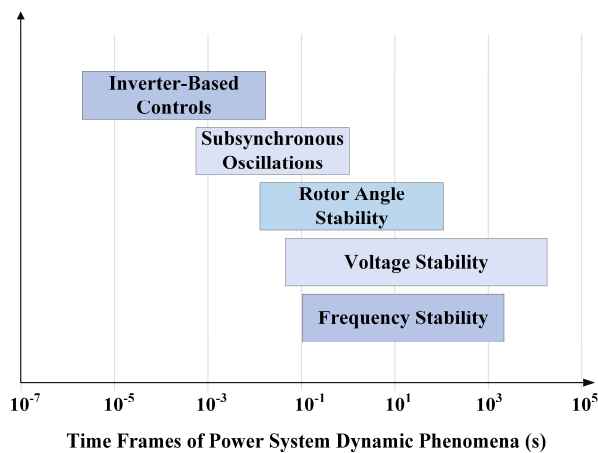


FIGURE 4. Time frames of power system dynamic phenomena.

B. VOLTAGE STABILITY

In traditional power systems, voltage stability depends on the power system's capacity to maintain steady voltage levels across buses after disturbances [22]. Large-disturbance voltage stability is divided into short-term and long-term categories. Short-term stability is studied over several seconds and concerns components that experience rapid load changes. Long-term stability deals with slower-acting devices; the study period lasts several minutes. Since RES are interfaced with the grid using power electronic converters, voltage stability in RES-based grids is mainly influenced by the control mechanisms of these converters.

C. FREQUENCY STABILITY

Frequency stability refers to the power system's capability to maintain a constant frequency following significant imbalances between load and generation [21]. It has two subcategories: long-term (minutes) and short-term (seconds) stability. Inertia is essential in studying frequency stability in the context of increasing the existence of grid-forming (GFM) and grid-following (GFL) converters. The frequency stability definition is extended to cover virtual and direct inertia. Direct inertia is a result of the energy stored in rotating generators, which helps them maintain rotation. Virtual inertia, on the other hand, replicates the inertia effect of a synchronous generator without relying on a rotating mass.

D. SUB/SUPER-SYNCHRONOUS OSCILLATIONS

Sub-synchronous and super-synchronous oscillations refer to system oscillations that occur below and above the fundamental frequency, respectively. According to the IEEE2800 standard, in a system with a fundamental frequency of 50 Hz, sub-synchronous oscillations typically occur within the range of 5 Hz to 55 Hz [24]. Other studies in literature have indicated that super-synchronous oscillations occur within the range of 50 Hz to 100 Hz [25]. These oscillations arise due to interactions between the generator (either a power electronics-based or rotor-based

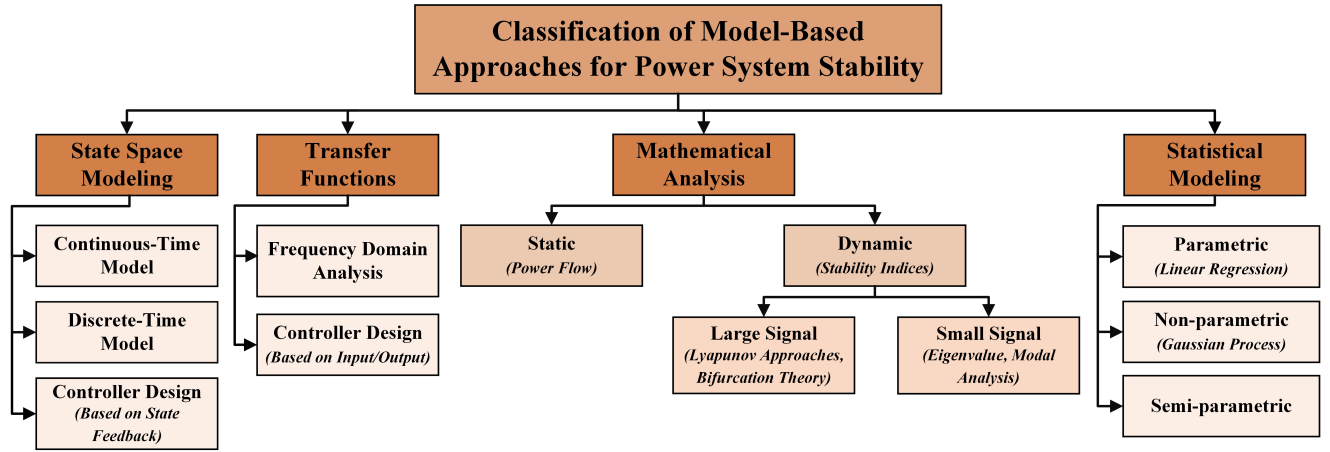


FIGURE 5. Classification of model-based approaches used for PSS analysis, based on a review of the literature.

generator) and the network or the interaction between two generators. They are distinct from electromechanical oscillations, which occur at much lower frequencies (0.5 – 3 Hz) and occur due to dynamic interactions of synchronous generators [26]. The study of sub/super-synchronous oscillations is particularly significant in the area of renewables due to RESs dependency on power electronic converters to be interfaced with the grid. For more details, the following reference is recommended for further reading [27].

E. HIGH RESONANCE FREQUENCY

The term "high resonance frequency" describes oscillations that emerge from the broad controller timeframe of IBRs, which can induce high-frequency oscillations throughout a wide frequency range. The high-frequency resonance range typically ranges from 300 Hz to 2000 Hz [28]. The main cause for the majority of high resonance frequency incidents is due to the operation of IBRs in areas with poor system strength and increased grid impedance. This reference is included for further reading on the subject [29].

III. MODEL-BASED APPROACHES FOR PSS

Researchers predominantly utilized model-based approaches to study PSS analysis in systems with extensive RES penetration. These approaches are generally classified into four main types: state-space modeling, transfer functions, mathematical analysis, and statistical modeling, as illustrated in Fig. 5. This section will explore the various model-based approaches found in the literature and categorize them accordingly. Table 1 summarizes the model-based approaches, focusing on stability, RES penetration, the networks tested, and the simulation tools used.

A. STATE SPACE MODELING

Based on findings in literature, state-space modeling methods for the application of PSS with RES integration can be generally classified into three main categories: continuous-time state-space models, discrete-time state-

space models, and controller design approaches based on state feedback.

Continuous-time state space modeling includes a set of first-order differential equations to represent how a certain system changes with respect to time. Several works focused on using dynamic equations for PSS analysis [30]. For example, Ref. [31] intended to investigate the transient synchronization behavior of hybrid power systems consisting of different GFM inverters, including synchronous generators and droop-controlled inverters. This was done by modeling the system under study using a second-order motion equation analogous to the traditional swing equation. A different approach based on statistical modeling was applied in [32], which utilized stochastic differential equations (SDE) theory to evaluate how stochastic excitation affects the small-signal PSS.

Other papers focused on studying PSS using or extending the traditional swing equation, as defined in (1):

$$\frac{2H}{\omega_{syn}} \omega_{pu}(t) \frac{d^2 \delta(t)}{dt^2} = p_{mp.u.}(t) - p_{ep.u.}(t) \quad (1)$$

Where δ is the power angle, H is the normalized inertia constant, ω_{syn} is the synchronous electrical radian frequency, $\omega_{pu}(t)$ is the per-unit electrical frequency, $p_{mp.u.}$ is the mechanical power, and $p_{ep.u.}$ is the electrical power. For instance, Ref. [33] employed a Hamiltonian Surface Shaping and Power Flow Control (HSSPFC) method, which is a two-stage approach based on Hamiltonian mechanics and was used to design feedback controllers for the RESs to improve the transient stability of the power system. Ref. [34] proposed an optimal placement method for virtual inertia in power systems. This was achieved by introducing virtual inertia control loops for the wind turbine generator (WTG) and solar PV models to improve frequency stability. The

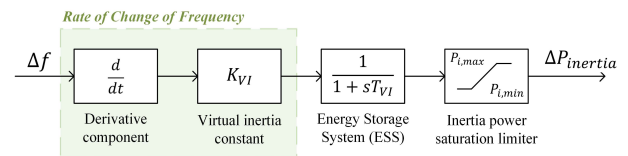


FIGURE 6. General control loop of virtual inertia.

system frequency dynamics were modeled using the swing equation,

Virtual inertia was also employed in [35], which addressed the challenges of increased RES penetration in microgrids by introducing a reinforcement learning-based virtual inertia control approach. The presented virtual inertia control diagram is shown in Fig. 6.

Other uses of state space modeling were also identified in several research papers. For example, Ref. [36] presented a method for assessing frequency stability in power systems with a significant percentage of RESs. The technique involved state space modeling of the system frequency response (SFR) model. A synchronous motor-generator pair (MGP) was presented in [37] to preserve transient stability in power networks with a high penetration of RESs. The proposed method employed state equations to model synchronous machines for small signal stability, considering different rotor electrical speeds and wind generation.

Similar to continuous-time modeling, discrete-time state space modeling describes the system behavior using difference equations rather than differential equations [38], [39]. Ref. [39] proposed a novel discrete-time state-space construction approach for eigenvalue analysis in SSO studies of AC/DC hybrid power systems with renewable integration. Compared to conventional continuous-time approaches, the method simplifies the development of the state matrix by discretizing the continuous-time state-space models of each system component and using equivalent circuits. The proposed approach was validated in a case study from northwest China, where the method was used to analyze the eigenvalues for SSO, and the results closely matched those of time-domain simulations in PSCAD/EMTDC.

On the other hand, state-space modeling can be used for controller design. Controller design based on state feedback includes using the system's states to calculate control inputs to stabilize the system as desired. For instance, Ref. [45] utilized state feedback control to reduce the frequency change rate to improve overall PSS in each area of a two-area power system with high RES penetration. Ref. [46] addressed the main challenges of achieving global asymptotic stability (GAS), voltage regulation (VR), and power regulation (PR) by employing a Lyapunov-based decentralized control (LBC). The feedback control approach aimed to improve the power system's transient stability while achieving VR and PR. Ref. [47] proposed an interconnection and damping assignment-passivity-based control (IDA-PBC) approach to improve transient stability and frequency and voltage regulation after the power system encounters a significant disruption. The control design for the IDA-PBC was based on state-feedback control law.

B. TRANSFER FUNCTIONS

Transfer functions are used to represent a relation between the input and output of a linear time-invariant (LTI) system. The application of transfer functions for PSS with RES integration can be broadly divided into two areas based

on findings in literature: frequency domain analysis and controller design.

For frequency domain analysis, transfer functions are important as they can be used to show how a system responds to inputs at various frequencies [40]. Ref. [41] presented a new approach based on the impedance network matrix for small-signal stability analysis of multiple-grid connected converter systems. The approach defined certain stability indices to quantify system stability and introduced an equivalent open-loop transfer function to simplify stability assessment. Frequency-domain analysis and time-domain simulations on a seven-converter system were used to validate the proposed approach. Super-synchronous oscillations in converter-dominated power systems were examined in [42], with a particular emphasis on grid-connected voltage source converters (VSCs) of WTGs in the current control timescale. The study developed a micro-grid model based on transfer function matrices to analyze system stability using a multivariable frequency domain method. Time domain simulations in Matlab/Simulink were used to validate the results.

Another important application of transfer function modeling is control design, which aims to determine a suitable controller transfer function that guarantees the system operates as desired. An improved stability control method for energy storage converters was presented in [43] in order to reduce oscillations in power systems with RES integration. All controllers were represented as transfer functions. The introduced approach employed the power impedance/admittance concept to improve systems stability by combining outer-loop power decoupling control with inner-loop virtual impedance-based active damping control. To validate the effectiveness of the proposed technique in reducing oscillations in RES dominated power systems, a renewable power system with an energy storage unit was modeled in PSCAD/EMTDC. Ref. [44] introduced a data-driven method for fine-tuning power system stabilizers to enhance electromechanical oscillation damping in modern power systems with increased RES penetration. The method eliminated the need for accurate power system models by using the virtual reference feedback tuning technique to redesign the numerator of the power system stabilizer transfer function using input-output measurements and a basic reference model. The proposed approach was verified through simulation of a Brazilian 7-bus test system.

C. MATHEMATICAL ANALYSIS

Mathematical analysis approaches are used to provide a framework for problem-solving through the study of continuous functions and other related theories. Mathematical analysis techniques for the application of PSS with RES integration can be broadly divided into two categories: static analysis and dynamic analysis, according to findings in the literature. Large-signal analysis and small-signal analysis are two more subcategories of dynamic analysis.

The steady-state analysis of PSS without any transient disturbances is known as static analysis. Techniques such as voltage stability assessment and power flow analysis are commonly used in this analysis. In [48], the continuous power flow method was utilized to obtain Power-Voltage (P-U) curves for static voltage stability analysis. Other papers used iterative numerical method approaches in simulation to solve nonlinear power flow equations that describe the grid's steady-state operation—this includes the Newton-Raphson (N-R) method [49], [50], and iterative power flow [51].

The study of how systems change over time in response to disturbances is known as dynamic analysis, and it is divided into two subcategories: large-signal and small-signal analysis. Large-signal dynamic analysis approaches – such as Lyapunov-based approaches and methods based on bifurcation theory – deals with analysis of how the system responds to major disturbances. Lyapunov-based approaches were utilized in several different proposed methods. Ref. [52] proposed an algebraic equation using sufficient robust frequency stability conditions, with the main aim of presenting a frequency stability framework that will allow for easy integration of optimization algorithms. This was achieved using a nonlinear average electrical grid model and Lyapunov's direct method extended by perturbation analysis to obtain the proposed algebraic equation. Other methods utilize Lyapunov theory for statistical-based approaches. For instance, Ref. [53] employed the structure-preserving transient energy function for the stochastic Lyapunov stability analysis of transient stability. A technique for angle stability analysis, considering cascading failures in power systems under various operating conditions, was proposed in Ref. [54]. Flow transfer theory and matrix of transfer probability between several operating conditions were obtained to represent the system operating conditions. The Lyapunov function containing the discrete Markov power system model for this application was then derived and used to define the robust stochastic stability criterion. On the other hand, a bifurcation approach was presented in [55] to evaluate PSS with high PV penetration while taking net bus load forecast error correlations into account. Testing on IEEE 3-bus, IEEE 33-bus, and IEEE 123-node systems revealed that the proposed technique enhanced voltage stability. Matlab was used to perform simulations.

Contrary to large-signal dynamic analysis, small-signal dynamic analysis studies a system's response to minor disturbances. Common small-signal analysis approaches include eigenvalue analysis and modal analysis. By computing the eigenvalues of a system's matrix, eigenvalue analysis can be used to assess a system's stability. In modal analysis, the natural modes of oscillation of the system are examined to help in identifying particular dynamic behaviors. While modal analysis focuses on analyzing the system's dynamics – such as frequency, damping, and oscillatory behavior – eigenvalue analysis is only a mathematical step in modal analysis that computes eigenvalues to evaluate system stability. Eigenvalue analysis was found in several proposed approaches in the literature.

For instance, Ref. [56] developed a two-step heuristic approach that employed multidimensional parameter variation to quantify voltage stability limits in power systems. A small-signal stability analysis method for large-scale power networks with a high penetration of IBRs – such as GFM and GFL inverters – was proposed in [57]. The suggested method used the IBR black-box models to simplify and improve the precision of the PSS analysis. Based on that, the network transfer function matrix (NTFM) was utilized to represent the complete power system network, and the sequence impedance and admittance models for the black-box IBRs were found using the small-signal perturbation approach. Perturbation theory, which is closely related the eigenvalue analysis, was employed in [58]. This paper focused on assessing robust small-signal stability region (RSSR) to preserve system stability, even with the interference of uncertain perturbations.

On the other hand, modal analysis was also a widely used method in literature for PSS studies. Ref. [59] introduced a frequency-domain modal analysis approach to address the problem of new oscillatory stability concerns because of the increased utilization of RESs. Ref. [60] proposed a discrete-time domain modal analysis approach to analyze power systems' SSO. The stability was evaluated by deriving the discrete-time system state matrix and obtaining its discrete-time eigenvalues.

Another essential dynamic analysis tool for studying PSS is using stability indices, which are indicators used to assess a power system's stability under disturbances. Stability indicators can be classified as either large-signal or small-signal dynamic analysis, depending on the particular index and application. Examples of well-known stability indicators include the synchronizing torque coefficient indicator and RoCoF. The system's ability to sustain rotor angle synchronism is indicated by the small-signal stability indicator known as the synchronizing torque coefficient, which was used in [61] to analyze the effect of different control modes on a radial power system. The stability index was evaluated for different PV penetration levels to find which control mode provides the most stability. The RoCoF is a large-signal stability index that refers to how quickly a power system's frequency changes over time, and it indicates how tolerant a system is to disturbances. Ref. [62] uses the RoCoF to evaluate the effectiveness of different frequency response approaches for enhancing frequency stability in power systems with high RES penetration. Similar to RoCoF, the rate of change of voltage (RoCoV) is used to quantify how quickly voltage changes over time to assess voltage stability. RoCoV was used in [63] to assess DC voltage stability in a DC microgrid. Other large-signal stability indices were also used in [64] and [65]. Ref. [64] addressed the issues of traditional approaches for assessing power system transient stability by introducing an approach to monitor and simulate post-disturbance voltages and obtain voltage-based indicators for transient instability. Ref. [65] aimed to assess PSS by deriving dynamic models and using them to conduct numerical simulations. Several stability

indicators were defined to evaluate the transient stability of the power system. Other small-signal stability indices were used in [66], which aimed to study the effect of network structures on small disturbance voltage stability (SDVS) by proposing an index that specifies which network structures are stable for different operating conditions.

D. STATISTICAL MODELING

Another important subcategory of mathematical modeling is statistical modeling, which is typically used for sample data generation and making predictions, and was widely used in previous works. Statistical modelling techniques for the application of PSS with RES integration can be broadly divided into three groups based on discoveries in the

TABLE 1. Summary of mathematical modeling methods used in literature for PSS.

	<i>Model-Based Method</i>	<i>Ref.</i>	<i>Stability</i>	<i>% of RE Penetration</i>	<i>Network</i>	<i>Simulation / Validation Tool</i>
State Space Modeling	Dynamic Equations	[30]	Frequency stability	“High”	- Northwest power grid	N/A
		[65]	Synchronous stability (<i>transient</i>)	N/A	- WSCC 9-bus - IEEE 39-bus	- Matlab - MATPOWER
		[31]	Synchronous stability (<i>transient</i>)	N/A	- WSCC 9-bus system	- Hardware-in-the-loop (HIL)
	Swing Equations	[34]	Frequency stability	“High”	- Eskom Western Transmission network	- DigSilent PowerFactory
		[33]	Synchronous stability (<i>transient</i>)	N/A	- One machine infinite bus (OMIB)	N/A
	State Space Modeling	[36]	Frequency stability	20%-60%	- Simplified northwest power grid system	N/A
		[37]	Synchronous stability (<i>transient</i>)	50%	- Single machine infinite bus test system - 4-machine 2-area system	N/A
Transfer Functions	Transfer Function Modeling	[40]	Small-signal stability	N/A	- Hybrid power generation/energy storage system	N/A
Analysis	Eigenvalue Analysis	[57]	Small-signal stability	20% to 90%	- Modified IEEE 14-bus - Reduced 240-bus WECC system	- PSCAD/EMTDC
		[59]	Oscillatory stability	“High”	- Simple passive circuit - Guyan wind power system	- PSCAD/EMTDC
		[60]	Subsynchronous oscillations	N/A	- Single wind farm integrated system - Multiple wind farms integrate AC/DC hybrid systems based on practical Hami, China systems.	- PSCAD /EMTDC
		[70]	Voltage stability	“Large”	- IEEE24-bus - IEEE118-bus	N/A
	Continuation Power Flow (CPF)	[52]	Frequency stability	N/A	- IEEE 39-bus test system	- DigSILENT PowerFactory
	Lyapunov Based Approaches	[64]	Synchronous stability (<i>transient</i>)	N/A	- New England 10-machine 39-bus power system	- Central Research Institute of Electric Power Industry (CRIPI)’s power-system dynamics analysis program
	Stability Indices	[66]	Voltage stability (<i>small disturbance</i>)	0% to 95%	- Hong Kong power grid - IEEE 14-bus, 25-bus, 57-bus, 118-bus systems	N/A
	Structured Perturbation Theory	[58]	Small-signal stability	“Large scale”	- Modified two-area system - New England 39-node system	- Matlab

literature: parametric, semi-parametric, and non-parametric techniques. In Table 2, the statistical modeling approaches found in the literature are summarized.

Parametric statistical modelling, which is used to estimate relationships or make predictions, makes the assumption that data follows a particular distribution that is determined by a few parameters. Examples of parametric modeling approaches utilized in literature include linear regression and Markov modeling. Linear regression was used in [67] to enhance stability in island grids with significant RES penetration. The paper introduced a unit commitment model with a data-driven frequency nadir constraint. The proposed approach surpasses minimum inertia limitations with negligible cost increases and reduces frequency variations during outages using a linear regression model. Ref. [68] employed Markov modeling in a technique to maximize the size and capacity of battery energy storage systems in PV-integrated distribution systems. Optimization complexity was decreased by modelling PV powers and loads as Markov processes.

On the other hand, non-parametric modelling employs adaptable techniques to describe patterns straight from the data without making any assumptions about the distribution of the data. Gaussian processes are an example of non-parametric modeling approaches. An approach based on the Gaussian process was presented in [69], which introduced a probabilistic framework based on the parallel partial Gaussian process (PPGP) for analyzing the transient stability of large probabilistic power systems.

Semi-parametric modelling combines non-parametric and parametric methods. While certain components of the model (parametric) are left undetermined, other parts (non-parametric) take on a specific form. Most work in literature regarding statistical modeling based methods for PSS with high RES integration utilize semi-parametric modeling, particularly methods that incorporate the stochastic response surface method (SRSRM) and Monte Carlo simulation. For example, Ref. [70] presented a unique method for examining how unknown power injections affect evaluations of voltage stability. The proposed approach first used the SRSRM

TABLE 2. Summary of statistical modeling methods used in literature for PSS.

<i>Statistical Modeling Methods</i>	<i>Probability Distributions</i>	<i>Ref.</i>	<i>Stability</i>	<i>% of RE Penetration</i>	<i>Network</i>	<i>Simulation Tool</i>
- Stochastic response surface method (SRSRM) - Latin hypercube sampling (LHS)	- Normal distribution: wind power	[70]	Voltage stability	"Large"	- IEEE24-bus - IEEE118-bus	N/A
- Global sensitivity analysis (GSA) - SRSRM	- Normal distribution: variability of the direction of active load increase - Beta distribution: solar irradiance	[71]	Voltage stability	"Significant"	- IEEE-9 bus - IEEE-118 bus	- Matlab
- Monte Carlo simulation	- Beta distribution: wind power	[73]	Voltage stability	"Large"	- IEEE 30-bus	N/A
- Monte Carlo simulation - Hierarchical clustering	- Normal distribution: uncertainties - Beta distribution: PV generation - Weibull distribution: wind generation	[75]	Synchronous stability (transient)	20%, 60%	- Modified IEEE 68 bus, 16-machine system - New England Test System and New York Power System	- DigSILENT PowerFactory
- Monte Carlo simulation	N/A	[74]	Voltage stability	20%, 50%, 80%	- IEEE 14 bus test system - IEEE 30 bus	N/A
- Lyapunov-based discrete Markov model	N/A	[54]	Synchronous stability	N/A	- IEEE 16-machine 68-bus New England-New York inter-connected system	- Matlab
- Parallel partial Gaussian process (PPGP)	N/A	[69]	Synchronous stability (transient)	N/A	- Texas 2000-bus system	- Matlab
- Linear regression	N/A	[67]	Frequency stability	"High"	- Maui Island grid	- PSS/E
- Monte Carlo simulation	- Gaussian distribution: solar irradiance - Weibull distribution: wind speed	[72]	Small-signal stability	"High"	- Modified IEEE 14-generator benchmark model (simplified Australian network)	- DiGSILENT/PowerFactory

approach to determine if the uncertainty of power injection could result in a voltage collapse. Next, a scenarios screening method was used to screen and rank the severe scenarios resulting in a voltage collapse. The SRSM approach was also used in [71], which employed a global sensitivity analysis (GSA) method to prioritize the RES variabilities affecting the power system's voltage stability. The proposed approach first presented a probabilistic model for load margin calculation considering RESs, then applied GSA to evaluate the impact of random input variables.

Other papers employed semi-parametric modeling using the Monte Carlo simulation method [72]. For instance, Ref. [73] proposed an approach for examining the impact of a high percentage of RESs on the system's Total Transfer Capability (TTC). Ref. [74] utilized the Monte Carlo simulation approach to construct an outline for voltage stability assessment while accounting for PV power generation and load demand uncertainties. Ref. [75] presented a probabilistic technique for evaluating transient stability in the context of a high RES penetration level. The optimal power flow (OPF) problem was solved to dispatch conventional generators, and Monte Carlo simulations were carried out for different networks and RESs penetration levels.

IV. OPTIMIZATION-BASED APPROACHES FOR PSS

Optimization is another essential approach for addressing the PSS challenges introduced by a high integration of RESs. The literature findings divide optimization methods into two subsections: metaheuristic algorithms and mathematical techniques. The different optimization-based methods that have been identified in literature for PSS with high RES penetration are summarized in Fig. 7. This section will explore those techniques and categorize them accordingly.

A. MATHEMATICAL OPTIMIZATION

Mathematical optimization focuses on finding an optimal solution for a specific problem by maximizing or minimizing objective functions with constraints. Mathematical optimization techniques have been employed in various PSS problems, and can be generally divided into three categories based on findings in literature: OPF methods, linear optimization, and non-linear optimization. Table 3 summarizes the findings of mathematical optimization methods. It displays each technique and its corresponding reference, stability analysis, RESs penetration level, and used network.

OPF approaches identify ideal operating conditions for power systems by optimizing certain objectives while making sure all system constraints are met. It should be noted that the methods based on OPF – including Dynamic OPF (DOPF), Reactive OPF (ROPF), and Transient Stability-Constrained OPF (TSCOPF) – are based on power flow or balance equations. Ref. [76] proposed a DOPF model to investigate the impact of RESs on static voltage and phase angle stabilities. The main objective of the DOPF was to minimize energy generation costs. Ref. [77] introduced an

ROPF model considering short-term voltage stability constraints. The ROPF used the supplementary control of devices, including synchronous condenser allocation and switchable capacitor groups. Furthermore, Ref. [78] suggested using a TSCOPF model to determine the system's ideal operating point while considering the transient stability of a power system with high RES integration. It investigated how transient stability, power dispatch, and overall generation cost are affected by inertia emulation. These parameters were studied for the cases of normal OPF, TSCOPF without inertia emulation, and TSCOPF with inertia emulation.

Linear optimization focuses on optimizing a linear objective function under linear equality and inequality constraints. Linear programming (LP) has been used in literature for assessing PSS for both traditional grids and grids with RES. In [79], LP solved the optimal coordination problem of directional overcurrent relays (DOCRs) used for overcurrent protection in traditional power systems.

On the contrary, non-linear optimization finds the optimal solution in problems with nonlinear objective functions or constraints. SOS programming was utilized in [80], presenting a direct technique for evaluating the transient stability of power systems containing RESs susceptible to tripping due to low voltage ride-through (LVRT) constraint violations. The Lyapunov functions and SOS programming were used to predict constrained stability regions (CSR) for non-linear switching systems to detect unstable fault-clearing times.

B. METAHEURISTIC OPTIMIZATION

A metaheuristic algorithm is a tool that rapidly solves complex optimization problems by iteratively searching for a sufficient near-optimal solution. Metaheuristic approaches can be classified into three main categories: population-based methods, trajectory-based methods, and hybrid methods. Typically, these approaches were used in literature

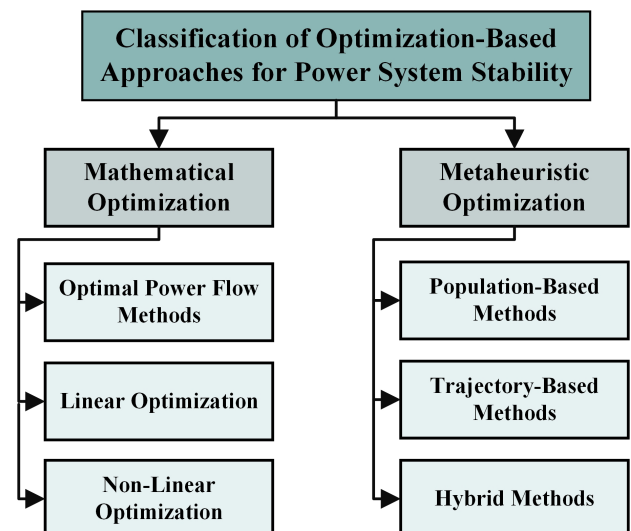


FIGURE 7. Classification of optimization-based approaches used for PSS analysis, based on literature review.

TABLE 4. Summary of metaheuristic algorithms used in literature for PSS.

<i>Metaheuristic Algorithms</i>	<i>Controlled Parameters</i>	<i>Type of Controller</i>	<i>Ref.</i>	<i>Stability</i>	<i>% of RE Penetration</i>	<i>Network</i>	<i>Simulation Tool</i>
- Donkey and smuggler optimization (DSOT)	- Tilt integral-derivative with filter (TIDF) - Fractional order PID (FOPID)	- TIDF regulator	[81]	Frequency stability	N/A	- Two-area nonlinear power system (TANLPS)	- Matlab/Simulink
- Grey wolf optimization (GWO) - Particle swarm optimization (PSO)	- PI	- Optimum demand response (DR) capacity for frequency control	[82]	Frequency stability	20%, 35%, 50%	- Single machine with centralized load model - Single area power system	- Matlab
- Modified particle swarm optimization with genetic algorithm (MPSOGA)	- TIDF → for load frequency control (LFC) - TID → for superconducting magnetic energy storage (SMES)	- Cooperative control of LFC and VIC - VIC: SMES based-TID - LFC: TIDF based	[85]	Frequency stability	“High”	- Two-area power system	- Matlab / Simulink
- Tabu search - Differential evolution (DE)	N/A	N/A	[84]	Voltage stability	N/A	- IEEJ EAST 10-machine system model	- Matpower - Developed with C++
- Particle swarm optimization-gravitational search algorithm (PSO-GSA)	- PI – PD cascaded control	- Cascaded control for automatic load frequency control (ALFC) application	[86]	Overall system	N/A	- Two-area multisource hybrid power system (HPS)	N/A
- Chimp optimization algorithm (ChOA)	- Linear gain (L) - Quadratic gain (Q) - Regulator gain (R)	- Model predictive control and linear quadratic Gaussian (MPC-LQG)	[83]	Frequency stability	“High”	- Single area power system - Two-area power system	- Matlab/Simulink

for optimally tuning control parameters. Table 4 summarizes the findings by displaying each reference and its respective metaheuristic method applied for tuning control parameters, the controlled parameters, and the type of controller used.

Population-based optimization approaches include preserving and evolving a population of possible solutions to a problem rather than enhancing a single solution. To achieve frequency stability, tilt integral derivative filter (TIDF) control was used in [81] for a two-area nonlinear power system (TANLPS). The TIDF controller parameters were adjusted using the donkey and smuggler optimization technique (DSOT), a newly developed metaheuristic method. Similarly, Ref. [82] Introduced an equivalent load frequency control (LFC) test model based on the Turkish grid and separately utilized the metaheuristic algorithms particle swarm optimization (PSO) and grey wolf optimization (GWO) to tune the optimum demand response (DR) capacity controller to enhance system frequency. The LFC model was modeled as a multi-source, single-area power system. Ref. [83] proposed a new control approach that combined model predictive control (MPC) along with the linear quadratic Gaussian (LQG) method for improving the frequency stability of the power system and LFC problems for systems subject to nonlinearities, high RESs penetration, and uncertainties. The introduced robust controllers were tuned using the Chimp organization algorithm (ChOA), a metaheuristic algorithm.

Trajectory-based optimization approaches iteratively follow a trajectory through the search space in order to improve a single candidate solution. Ref. [84] utilized Tabu search – a trajectory-based optimization approach – along with differential evolution (DE) for optimal battery placement in an IEEJ EAST 10-machine system model to maximize voltage sensitivity in all power system nodes against any disruptions in RESs.

Using features from several optimization techniques, hybrid methods combine their advantages to more efficiently address complex problems. For instance, Ref. [85] proposed a hybrid metaheuristic method known as modified particle swarm optimization with genetic algorithm (MPSOGA) to cooperatively tune the load frequency controller parameters as well as the superconducting magnetic energy storage (SMES) based virtual inertia controller (VIC) parameters. Moreover, to analyze the stability of the proposed controller designed in [86], the Hankel method was applied to the derived system state space model for model order reduction. The paper proposed a hybrid particle swarm optimization-gravitational search algorithm (PSO-GSA) technique for tuning a cascaded control system for automatic load frequency control (ALFC) application.

V. AI-BASED APPROACHES FOR PSS

Artificial Intelligence (AI) has become an important tool in modernizing power systems, especially in the context of

high RES penetration. More specifically, Machine Learning (ML), a branch of AI that utilizes past data to detect patterns and generalize information, has been extensively applied to this particular application. In Fig. 8, an overview is presented of ML approaches used in the literature for stabilizing power systems with high RES penetration, which have been divided into four main categories: supervised learning, unsupervised learning, reinforcement learning, and deep learning. Brief definitions of the mentioned ML methods are presented in Fig. 9, while Table 5 summarizes their applications as documented in the literature.

A. SUPERVISED ML

Training models on labelled data to predict results for new inputs is known as supervised ML. It was noted that supervised learning methods were more commonly used than unsupervised learning methods for PSS analysis with high RES penetration, particularly in applications such as stability forecasting. Classification and regression are the two primary categories into which supervised ML techniques fall into. Supervised learning approaches can be used for both classification and regression. Methods typically used in literature for the application of PSS with high RES penetration include the support vector machine (SVM), ensemble methods, neural networks, and decision trees (DTs).

SVM is a frequently used supervised learning approach that separates data into classes by determining the most appropriate boundaries between the data. SVM has been widely used as a classification tool. For instance, Ref. [87] employed an SVM classifier using measured synchrophasor data first to forecast the voltage stability state of the power system. The prediction results triggered the designed control MIMO MPC strategy to support control variables and maintain transient voltage stability. Additionally, Ref. [88] presented a convolutional neural network (CNN) approach

based on SVM to classify and assist in transient stability assessment in power systems.

SVM has also been used for regression tasks. For instance, SVM was employed to handle regression tasks in [79], which proposed an online monitoring approach to assess power system voltage stability. This was done using training data from phase measurement units (PMUs) to train the Weighted Least Square SVM (WLS-SVM) approach to acquire and track the real-time voltage stability index.

Another supervised learning tool is ensemble methods, which combine several different ML techniques for an improved algorithm. Ensemble methods have been commonly used for classification. For instance, Ref. [89] presented a transient stability assessment method based on two supervised ML methods: extreme gradient boosting (XGBoost) and factorization machine (FM). The XGBoost approach was used for feature construction to transform the power flow into sparse matrix form, while FM was used as a fault classifier. The training data was obtained by time domain simulation using PSD-BPA. Python programming was used to implement the baseline ML models. In [90], a stacking ensemble classifier (SEC) was proposed for predicting grid stability in decentralized smart grids. The introduced model combines several supervised learning approaches – including XGBoost, Categorical boosting (Catboost), and Light Gradient Boosting Machine (LGBM) – to accurately classify grid instabilities. Thorough testing of the proposed method showed that the SEC outperforms other benchmark models. Ref. [91] developed a hierarchical intelligent system (IS) method to address transient voltage collapse classification and unacceptable dynamic voltage deviation prediction. To achieve this, predictor and classifier ensemble-based learning methods consisting of neural networks with random weights (NNRW) were used for offline training, real-time application, and online updating patterns.

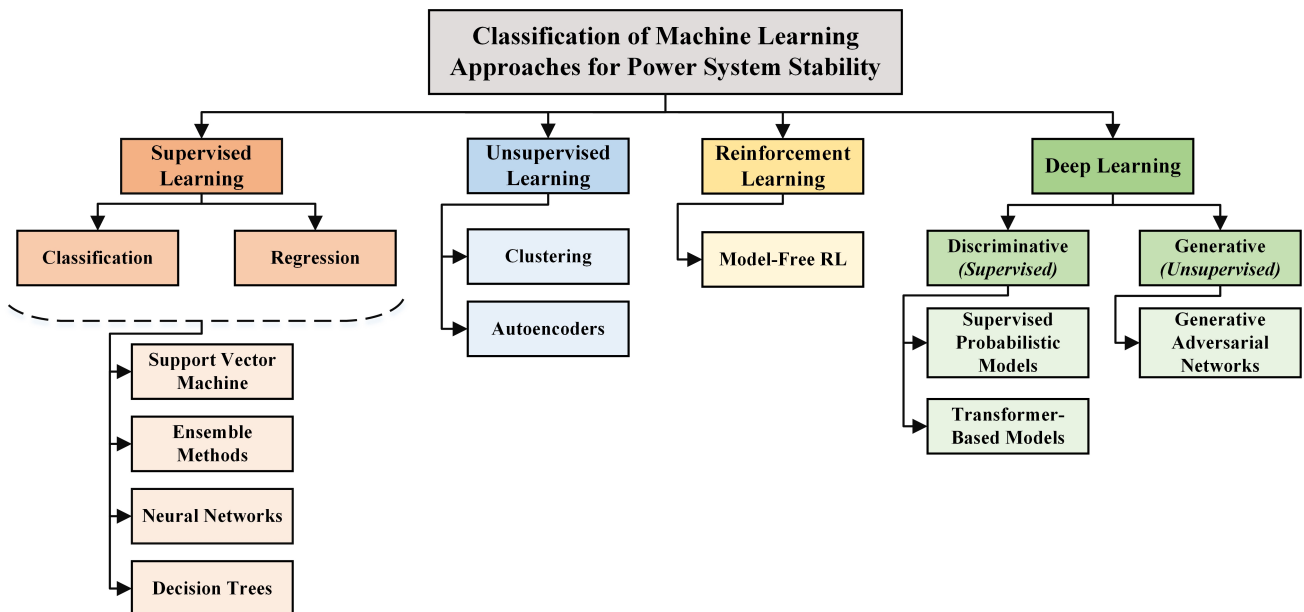


FIGURE 8. Classification of ML-based approaches used for PSS analysis, based on literature review.

Ensemble methods have also been employed in the context of regression. Ref. [92] proposed a novel ensemble method for regression by combining multiple ML methods for tuning PSS parameters to dampen undesired oscillations. Several ML approaches – including extreme learning machine (ELM), neurogenetic (NG) system, and multi-gene genetic programming (MGGP) – were combined using the backtracking search algorithm (BSA) based ensemble method for real-time PSS parameters tuning.

Neural network models work by imitating the human brain to make predictions based on learned patterns. Neural networks have been widely used in literature for classification. Long-Short-Term Memory (LSTM) is a type of neural network applied in [93], which presented an enhanced Bi-directional LSTM (BI-LSTM) approach for classification with a class-balanced focal loss function to guide the data-driven transient stability assessment model training. The main issue tackled was the disproportion of stable and unstable measured power system data, which may result in stable samples being prioritized more than unstable samples in the model training. Ref. [94] presented a novel classification method for improving ML-based PSS assessment techniques called the robust online stability assessment (ROSA) to safeguard models against adversarial scenarios that use white-box and black-box techniques. The proposed method included two independent neural network modules: a self-supervised learning-based module and a supervised learning-based module, which addressed primary stability analysis.

Neural networks were also used for regression tasks. Ref. [95] used the regression approach to add Lyapunov functions for transient stability analysis and determining post-fault equilibrium locations' stability region (SR). Stochastic gradient descent (SGD) was used to derive the Lyapunov function, and neural networks were employed as the learners of the Lyapunov function. Ref. [96] used regression approach by employing a learning-based method to provide a unique control strategy for PSS issues. Deep neural networks were used in the two-module structure of the suggested method to find a candidate Lyapunov function and rules of control.

DT is another widely used supervised ML approach that uses tree-like structures to illustrate decisions and their potential outcomes. DTs can be used for both classification and regression purposes. Ref. [97] proposed a strategy that utilized DTs for both classification and regression to predict multi-mode damping of power systems with RESs integration. Wide-area measurement systems (WAMs) were used to collect mode-damping data to be handled by DTs to indicate PSS. Probabilistic wind power and PV generation models were first defined to generate training data. The DT was constructed offline by utilizing the learning data and was regressed online to carry out its predictions.

B. UNSUPERVISED ML

Unsupervised learning is another important subcategory of ML. In unsupervised ML, models use pattern recognition to analyze and classify unlabeled data without predetermined outcomes. Based on findings in literature, unsupervised ML approaches used for the application of PSS with high RES integration can be divided into two: clustering techniques, and autoencoders. It should be noted that works typically use unsupervised learning approaches along with supervised learning methods.

Clustering approaches are used to put similar data points into groups according to common traits. For instance, [98] proposed a practical framework that first used a filter to identify worst-case scenarios from other possible operating conditions. Next, a clustering method was employed to select representative worst-case operating conditions to study power system frequency stability. The filter design utilized supervised ML methods – including random forest (RF), DTs, and artificial neural networks (ANN), to identify and classify the worst-case operating points of contingencies. After that, unsupervised ML was applied using a clustering process to select the representative worst-case operating point. Clustering by affinity propagation was used in [99], which presented a platform based on supervised and unsupervised ML approaches for time-series preliminary assessment of transient stability. Security-Constrained OPF (SCOPF) simulation obtained the

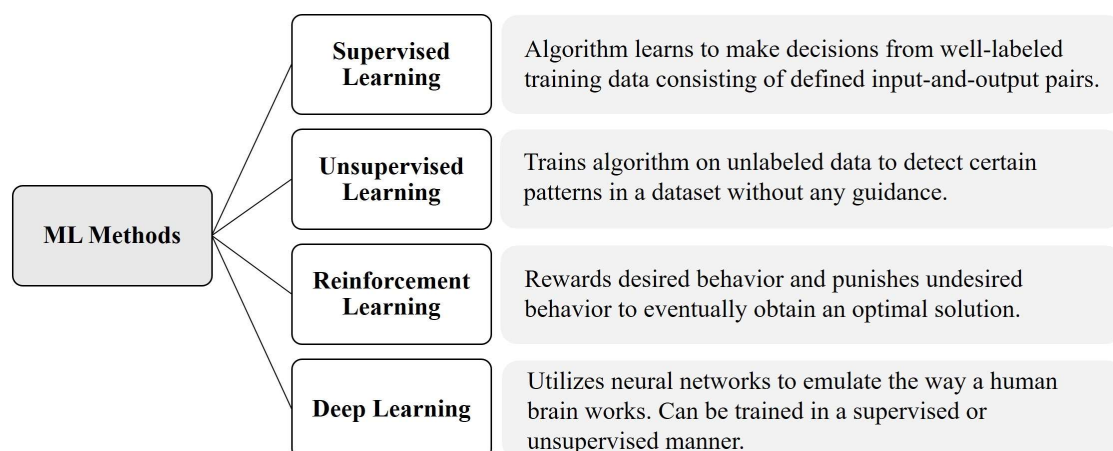


FIGURE 9. Definitions of main ML approaches used in literature for PSS analysis.

TABLE 5. Summary of AI-based approaches used in literature for PSS.

	<i>ML Method</i>	<i>Ref.</i>	<i>Stability</i>	<i>% of RE Penetration</i>	<i>Network</i>	<i>Simulation Tool</i>
Supervised Learning	- Support vector machine (SVM) classifier	[87]	Voltage stability (<i>transient</i>)	Above 20%	- Kundur two-area power system	- Matlab /Simulink
	- Random forest (RF)	[98]	Frequency stability	21%	- Chilean Northern Interconnected System (NIS)	- PowerFactory - Python
	- Decision trees (DTs)					
	- Artificial neural network (ANN)					
	- Class-balanced focal loss function (CBfocal) Bi-directional Long-Short-Term Memory (Bi-LTSM)	[93]	Synchronous stability (<i>transient</i>)	“High”	- New England 10-generator 39-bus system	- PSD-BPA
	- Weighted least square support vector machine (WLS-SVM)	[79]	Voltage stability	N/A	- New England 39-bus - IEEE 118-bus	N/A
	- Support vector machine-based convolutional neural networks (SVM-CNN)	[88]	Synchronous stability (<i>transient</i>)	N/A	- IEEE 14-bus test system	- Matlab
	- Extreme learning machine (ELM)	[92]	Overall stability	N/A	- SMIB with PSS - SMIB with PSS-UPFC	- Matlab
	- Gradient boosting regressors	[99]	Synchronous stability (<i>transient</i>)	0%, 25%, 35%, 45%	- IEEE-39 test system	- Python - MATPOWER - PSS/E
	- Factorization machine (FM)	[89]	Synchronous stability (<i>transient</i>)	N/A	- IEEE New England 10-machine 39-bus system - IEEE 16-machine 68-bus system - IEEE 47-machine 140-bus system	- Python - PSD-BPA
	- Extreme gradient boosting (XGBoost)					
	- Decision trees (DTs)	[97]	Overall stability	N/A	- 16-generator, 68-bus system of New England-New York interconnected power system	N/A
	- Neural lyapunov control	[96]	Synchronous stability (<i>transient</i>)	N/A	- 1-machine system - 3-bus system - 13-bus system - 123-bus system	N/A
	- Neural networks based on the Lyapunov direct method	[95]	Synchronous stability (<i>transient</i>)	N/A	- IEEE 9-bus 3-machine system	N/A
	- Least absolute shrinkage and selection operator (Lasso) regression	[100]	Synchronous stability (<i>transient</i>)	29%	- Modified IEEE 39-bus system	N/A
Unsupervised Learning	- Clustering technique	[98]	Frequency stability	21%	- Chilean Northern Interconnected System	- PowerFactory - Python
	- Neural networks with random weights (NNRW)	[91]	Voltage stability (<i>short-term</i>)	N/A	- New England 39-bus, 10-machine system	- PSS/E - Matlab
	- Clustering by weight-adjusted affinity propagation	[99]	Synchronous stability (<i>transient</i>)	0%, 25%, 35%, 45%	- IEEE-39 test system	- Python - MATPOWER - PSS/E
	- Stacked de-noising auto encoder (SDAE)	[100]	Synchronous stability (<i>transient</i>)	29%	- Modified IEEE 39-bus system	N/A
	- Principal component analysis (PCA)	[79]	Voltage stability	N/A	- New England 39-bus - IEEE 118-bus	N/A
Reinforcement Learning	- Lyapunov regularized reinforcement learning method	[101]	Synchronous stability (<i>transient</i>)	N/A	- IEEE New England 10-machine 39-bus	- Python - Matlab
	- Barrier-certificated power system reinforcement learning control	[102]	Synchronous stability (<i>transient</i>)	N/A	- IEEE 39-bus system with 10 machines	- Python
Deep Learning	- Transformers	[104]	Voltage stability (<i>short-term</i>)	N/A	- IEEE 39-bus	- Python - PSD-BPA
	- Deep sigma point process (DSPP)	[103]	Synchronous stability (<i>transient</i>)	21.3%	- IEEE 39-bus system - South Carolina 500-bus system	N/A
	- Generative adversarial networks (GANs)	[99]	Synchronous stability (<i>transient</i>)	0%, 25%, 35%, 45%	- IEEE-39 test system	- Python - MATPOWER - PSS/E

unlabeled training dataset. Dimensionality reduction on the dataset was then carried out using gradient-boosting regressors, a supervised ML approach. The unlabeled dataset was clustered using weight-adjusted affinity propagation, an unsupervised ML approach. Generative adversarial networks (GANs) – an unsupervised ML method – were then used to generate synthetic data to increase instances in predictive model training. Next, the supervised learning phase utilized gradient-boosting regressors to train the predictive models of the labeled datasets obtained for different scenarios.

An autoencoder is a type of neural network that compresses and reconstructs data to learn effective representations of the data. Ref. [100] presented a stacked denoising autoencoder (SDAE)-based probabilistic prediction approach for addressing uncertainties of RESs in transient stability assessment. This method aimed to learn the mapping relation of transient stability margin and operational features. The proposed model was implemented using the TensorFlow framework. To train the SDAE model, initial operation scenarios were created using Latin hypercube sampling (LHS), and the least transient stability index was obtained using time domain simulation.

C. REINFORCEMENT ML

Reinforcement learning (RL) is another ML branch that involves an agent learning decision-making skills through interactions with its environment. RL can be divided into different branches, but the branch most utilized in PSS studies for high RES penetration is model-free RL [35]. In model-free RL, an agent uses interactions with the environment to learn a policy without creating a model of the dynamics of the environment. Model-free RL was employed in [101], which presented a Lyapunov regularized RL method for optimal frequency control for enhancing transient stability, where the neural network controller was trained to punish behaviors that do not satisfy the Lyapunov conditions. Model-free RL was also used in [102], which proposed a general solution framework for conventional control problems using RL-based algorithms and neural barrier functions to maintain transient stability and satisfy safety constraints.

D. DEEP ML

Deep learning is a subcategory of ML that extracts representations and patterns from data by using multi-layered neural networks. Generally, deep learning is further divided into two main categories: discriminative (supervised) and generative (unsupervised) learning. Based on findings in literature for the application of PSS with high RES integration, common discriminative approaches include supervised probabilistic models, as well as transformer-based models. Moreover, frequently employed generative learning methods include GANs.

Supervised probabilistic models use labelled data to predict the probability of particular outcomes. An example of supervised probabilistic models includes the parametric Bayesian approach, which defines a set of parameters and

makes predictions by combining observed data with past knowledge. The parametric Bayesian approach was employed in [103], which introduced a deep sigma point process (DSPP)-assisted chance-constrained preventive control approach to address the uncertainty of RESs using transient stability analysis of power systems. DSPP, a parametric Bayesian approach, was used to predict the transient stability of the power system. The proposed method was implemented using the PyTorch framework. The primary goal of this study was to incorporate the DSPP probabilistic prediction model as a transient stability chance constraint into the TSCOPF in order to accelerate its calculation.

Unsupervised deep learning approaches have also been used in literature for PSS analysis in systems with high RES penetration. An example of unsupervised deep learning methods includes GANS, which is a ML model made up of two neural networks that work against one another. The generator produces artificial data, while the discriminator attempts to discern between the generator's fake and authentic data. Ref. [104] presented a novel approach using deep transfer learning based on PMU measurements for short-term voltage stability assessment (STVSA). The initial dataset used real-time dynamic data obtained by PMUs. Temporal ensembling was utilized to label samples, and least squares generative adversarial networks (LSGANs) were used for data augmentation (DA) to allow for effective deep learning on small-scale datasets. Furthermore, the transformer model – which is a supervised deep learning approach used for sequence-based problems – was proposed to increase the accuracy of the power system assessment.

VI. SIMULATION APPROACHES FOR PSS

Simulation is an essential tool frequently used in literature for analyzing PSS with high RES integration, but it is not typically used as a stand-alone approach. Instead, simulation approaches were used to verify and validate proposed PSS methods, whether model-based, optimization-based, or AI-based. Simulation methods are mainly divided into frequency and time domains. The time domain includes electromechanical and electromagnetic transient (EMT) simulation, while the frequency domain includes root mean square (RMS) simulation and phasor domain simulations. This section highlights how the different simulation methods were used to verify proposed approaches.

A. TIME-DOMAIN SIMULATION

Time-domain simulations are used to solve differential equations incrementally at each time interval to simulate how a system's variables change over time. Depending on the simulation type used, different dynamics can be captured. The most commonly found time-domain simulation approaches for PSS analysis with high RES integration include EMT, electromechanical, and dynamic simulations.

EMT simulation is a time-domain analytic method that solves detailed differential equations to accurately capture instantaneous electrical values. This allows for the study of

dynamic events in power systems and fast transients. Ref. [105] presented a practical guideline for EMT modeling to assist in preventing common errors and enhance simulation efficiency, accuracy, and stability. For further details on EMT modeling, [105] can be referred to. Other works in literature have employed EMT simulations to analyze and validate proposed methods. In [60], PSCAD/EMTDC was utilized to perform EMT simulations in order to validate the proposed discrete-time domain modal analysis method, which was employed to investigate SSO. Ref. [51] proposed a real-time control approach to enhance voltage stability with power systems with installed RESs. To validate the proposed method, EMT simulations were conducted using PSCAD on an IEEE 14-bus model. Ref. [59] employed EMT simulations to validate its proposed frequency-domain modal analysis for studying electromechanical oscillations in power systems with high RES penetration.

Electromechanical simulation is another time-domain analysis method that uses differential equations to capture the slow interactions between the electrical and mechanical components of the system. This type of simulation can capture the dynamic response of electrical and mechanical systems over longer time periods compared to EMT simulation. For further details, [7] explored the challenges of electromechanical stability in modern power networks with high RES penetration. The paper also emphasized the significance of understanding the interactions between electromechanical and electromagnetic dynamics in modeling and analysis. Electromechanical simulations have also been employed in other works. For instance, [106] investigated the stability characteristics of the Northwest Power Grid in situations where RES penetration was high. Using PSD-BPA software, an electromechanical grid simulation was created and used to evaluate the grid's synchronization, voltage, and frequency stability under various fault scenarios to examine the impact of changing the RES penetration from 40% to 70%. Ref. [107] presented a real-time frequency stability assessment method for power networks with high RES integration. A case study on a 50,000-node grid validated the proposed method, which involved modeling the system using an electromechanical simulation technique to evaluate transient and steady-state frequency margins.

Dynamic simulation refers to time-domain simulations that examine a system's time-varying behavior in response to major and minor disturbances. This is a more general time-domain simulation approach as it includes both EMT and electromechanical simulations. The performance of wind diesel isolated microgrids was modeled and analyzed using dynamic simulation in [108]. The paper offered a summary of dynamic simulation techniques, emphasized the value of high-order electrical machine models for accurate outcomes, and offered a useful Matlab/Simulink dynamic simulation example. Ref. [108] can be used for further reading on dynamic simulation in low-inertia isolated power systems. Dynamic simulations have also been used in other works. Using ML techniques and dynamic simulations, [99]

proposed a platform for evaluating transient stability in power systems with high RES integration. The required ML training dataset was obtained using the SCOPF simulation. Then, ML methods were carried out on the data using programming techniques, and dynamic simulation was done to label the dataset. Ref. [109] analyzed the transient stability of a hypothetical power system network through dynamic time-domain simulation. Different cases were simulated using Power System Dynamic software, including assessments of transient stability with increasing penetrations of distributed generation (DG) and RESs (9%, 40%, 80%, and 110%).

Numerical integration techniques are used in time-domain simulations to solve differential equations that characterize the dynamics of the system. By integrating the system's differential equations over small time periods, numerical integration techniques aim to determine the system's state at successive time points [105]. These methods are important for modeling transient behaviors in time-domain simulations, including frequency fluctuations and system stability following disturbances. The Euler method is a numerical integration approach that uses the slope at each time step to iteratively update values in order to estimate solutions to differential equations. This approach was used in [110], which presented a non-sampling time-domain simulation framework for evaluating the transient stability of power systems under stochastic continuous disturbance. It solved stochastic differential-algebraic equations using a Milstein-Euler prediction-correction iteration strategy, which provided faster convergence and higher accuracy than traditional numerical integration approaches. Matlab simulations were used in case studies on IEEE 39 and 118-bus systems to verify the accuracy of the framework. A more accurate numerical integration approach compared to Euler's method is the trapezoidal integration technique, which approximates the integral of a function by splitting it into trapezoids and adding up their areas. The trapezoidal integration method was employed in [111], which proposed a transient voltage security analysis approach that takes into account the impact of loads on power systems and the dynamic behavior of renewable energy plants. Differential-algebraic equations were developed to simulate the dynamics of the system. The study eliminated the necessity for Jacobian matrix computations by creating an enhanced implicit trapezoidal integration approach with polynomial interpolation correction technique to solve the equations effectively. In case studies on a modified IEEE 39-bus system, the approach showed great accuracy.

B. FREQUENCY-DOMAIN SIMULATION

To investigate steady-state behavior and harmonic responses, frequency domain simulation examines how a system responds to sinusoidal inputs at different frequencies, usually with the use of Fourier transforms. Although frequency-domain simulation approaches are not as widely used as time-domain approaches in the literature, they

remain an important tool for PSS analysis. The most frequently used frequency-domain simulation techniques for the purpose of PSS analysis of systems with high RES integration include RMS and phasor-domain simulation.

By excluding high-frequency transients, RMS simulations concentrate on the averaged behavior of voltages, currents, and power across a fundamental frequency cycle. Since it cannot capture high-frequency dynamics, this method is less accurate for applications where studying detailed transient impacts is critical. It should be noted that RMS simulations differ from electromechanical simulation, which captures detailed dynamics focusing on the interaction between electrical and mechanical components within the system. For further reading, [112] discusses the use of RMS simulations to predict and analyze transient responses to disturbances in large power systems. In the paper, EMT simulations are compared with RMS approaches. Other works in literature have also employed RMS simulations. For example, [113] used RMS simulations in DIgSILENT PowerFactory to study the stability effects of incorporating RES into the European power grid. Under different penetration levels (10%, 20% and 30%), the study investigated the dynamic behavior of the grid by substituting distributed static generators for conventional synchronous generators. Ref. [114] performed RMS simulations using DIgSILENT and Matlab to assess transient, frequency, and small-signal stability and examine the impacts of load modeling on various power system stabilities. Moreover, Ref. [75] proposed a probabilistic transient stability assessment method for power systems with high RES levels. The study used hierarchical clustering to identify the critical generators most likely to experience instability using RMS simulations in DIgSILENT PowerFactory.

Another important frequency-domain simulation tool is phasor domain simulations. Phasor domain simulation is suitable for slow dynamics like load flow and small-signal analysis since it assumes a constant frequency and employs steady-state sinusoidal signals represented as phasors. The primary difference between this approach and RMS simulation is that phasor domain simulation assumes steady-state conditions, while RMS simulation includes time-domain dynamics. Ref. [115] presented a review of the latest developments in OPAL-RT's ePHASORSIM tool for phasor domain simulations, used for applications such as real-time power system modeling. The study emphasized the advancements in power system testing, co-simulation, and model conversion, particularly when using RESs. In [116], the Maui power system was used for comparison between the performance of EMT and phasor domain modeling tools for power systems with high integration of IBRs. Despite the computationally lower cost of phasor domain tools, the study concluded that EMT simulations are required for more accurate modeling and representation of system dynamic in systems with large shares of IBRs.

VII. DISCUSSION

Despite the environmental benefits of increasing the integration of RESs in the electrical grid, this transition introduces significant challenges, such as fluctuating energy output due to unpredictability and low inertia, which can adversely affect PSS. This review was conducted to address the PSS issues that arise from the increasing presence of RESs. The paper provides a comprehensive overview of the methods used for PSS assessment, categorizing them into model-based, optimization-based, and AI-based approaches. A comparative evaluation of the different techniques was done based on the addressed PSS category, how the RES penetration was quantified, the simulation tools used, and the network topology considered.

To address the specific challenge of the stability issues related to increased RES penetration, a modified PSS classification was proposed. This classification divides stability into two categories: fundamental component stability and non-fundamental component stability. Fundamental component stability addresses classical PSS, focusing on power system instability around the operating system frequency. In contrast, non-fundamental component stability covers instability issues at frequencies higher or lower than the system's fundamental frequency.

Among the approaches reviewed, model-based methods were the most frequently used for PSS analysis, which is expected as they are considered the more traditional methods. These approaches were classified into four main categories: state-space modeling, transfer functions, mathematical analysis, and statistical modeling. State-space modeling approaches may provide a mathematical framework for analyzing the dynamic behavior of power systems, while transfer functions can often be employed in controller design to enhance system stability. Various analytical approaches in the literature offer additional tools for assessing system stability under RES penetration. Statistical modeling methods, often combined with other approaches, help account for the uncertainty introduced by RES integration.

Optimization-based approaches, while less frequently used than model-based methods, have a significant impact on enhancing PSS in grids with high RES penetration. These approaches are categorized into mathematical optimization and metaheuristic optimization. Mathematical optimization methods typically involved solving OPF problems to maintain system stability under increased RES penetration conditions, while metaheuristic optimization was mostly used for tuning controllers to enhance system stability and performance.

In recent years, AI-based methods have become essential to modern PSS analysis, particularly in systems with high RES penetration. The AI-based approaches in the literature are predominantly ML methods, categorized into supervised, unsupervised, reinforcement, and deep learning. Supervised learning is the most commonly used approach, mainly due to the large availability of labeled historical data, and was often applied to applications such as real-time stability assessment and control. Unsupervised ML was typically employed to

enhance data handling and predictive accuracy, while reinforcement learning and deep learning, though less common, contributed by optimizing control strategies and improving stability predictions. Additionally, many studies combine multiple ML methods into ensemble approaches to create more robust algorithms for enhancing PSS in systems with significant RES integration. Overall, AI-based methods were shown to have several benefits, including the ability to handle large amounts of data, and assist in making real-time predictive decisions based on identified patterns. When compared to conventional approaches, these methods provide better accuracy and scalability. However, there are still some notable challenges regarding AI-based methods – including the dependence on large training datasets, and the complexity that arises in generalizing these methods to be applied for different power system configurations.

Simulation is an important tool for implementing and verifying proposed approaches, supporting model-based, optimization-based, and AI-based methods in addressing the challenges posed by RES integration in power systems. Generally, simulation methods are divided into time-domain and frequency-domain approaches, where time-domain simulations include electromechanical, EMT, dynamic simulations, as well as numerical integration techniques. Frequency-domain simulations include RMS and phasor-domain simulations. This paper highlighted how these various simulation techniques were applied to validate proposed approaches for the application of PSS with high RES penetration, with Matlab/Simulink being the most popular simulation tool because of its versatile applications in control design, power system modeling, and time-domain simulation capabilities. PSCAD was also a widely utilized simulation software for modeling detailed interactions and accurately simulation electromagnetic transients and system dynamics. Despite being less widely used, DIGSILENT PowerFactory was utilized in some works for RMS simulation and dynamic stability studies.

Overall, the combination of model-based, optimization-based, and AI-based approaches provides a comprehensive framework for addressing the complex challenges introduced by the increasing penetration of RESs into power systems. Although this review provides a thorough overview of PSS approaches for high RES integration, it is important to note that there are still very few real-world practical applications of these techniques. Usually, only simulation case studies have been used to test the proposed methods in literature. The difficulty of deployment, fluctuating grid conditions, and requirements for great processing resources are all possible reasons for this gap in real-world applications.

VIII. CONCLUSION AND FUTURE WORK

In conclusion, this paper addressed the issues arising from the increasing RES penetration in the electrical grid by conducting a comprehensive review of current literature. A modified PSS classification for power systems with increased RES penetration was proposed. Moreover, a

variety of methodologies for stabilizing power systems with high RES penetration found in literature were systematically classified into model-based, optimization-based, and AI-based approaches. Model-based methods, the most traditional and widely used, provide a strong foundation for analyzing power system dynamics and designing stability-enhancing controllers. Optimization-based approaches, though less frequently employed, play a crucial role in improving PSS by solving complex optimization problems to enhance system performance under RES integration. AI-based approaches are increasingly utilized in modern PSS analysis, with supervised learning being the most prevalent due to the wide availability of labeled historical data. These advanced techniques offer powerful tools for real-time stability assessment, control optimization, and predictive analytics, particularly in systems where RES penetration introduces significant uncertainty.

Overall, incorporating these diverse model-based, optimization-based, and AI-based methodologies has the potential to create more robust frameworks for PSS analysis. This review paper aims to provide researchers in renewable energy with a clear and concise framework for evaluating PSS assessment techniques and highlighting promising research directions. To address the challenges that have been discussed, further studies should focus on particular research questions, such as “How can real-time stability assessment and control in high-renewable energy systems be enhanced by advanced AI techniques?” Additionally, investigating emerging technologies could also address the current PSS challenges faced due to high RES penetration. This includes investigating innovative power electronic advancements in grid-forming inverters and high-voltage direct current (HVDC) technologies to improve power systems’ ability to integrate and stabilize RESs, or digital twins for real-time monitoring and predictive maintenance. The aim of future work should be to improve the incorporation of novel techniques into practical power system applications.

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JANA SHEIKH ALI received the B.Sc. degree in Electrical Engineering from Qatar University in 2023. She is currently a Research Assistant at Qatar University. Her research interests include power systems and renewable energy.



YAZAN QIBLAWEY (M'18) received his B.Sc. degree in Electrical Engineering from Qatar University, his M.Sc. degree in Electrical and Electronics Engineering from the University of Nottingham, UK, and he is currently pursuing his Ph.D. in Electronic and Electrical Engineering at the University of Strathclyde, UK. He worked as a researcher with Qatar University between 2019 and 2021. He is currently a Power System Engineer at Iberdrola Innovation Middle East,

Qatar, where he focuses on system stability, converter interactions, and renewable energy integration. His research interests include power system stability, control of renewable energy systems and the use of machine learning for simulation and optimization. Yazan has been actively involved in several industry projects related to the integration of renewable energy into the grid, converter interactions, simulation of EMT systems, techno-economic optimization of renewables and storage integration.



ABDULRAHMAN ALASSI (*Senior Member, IEEE*) received the B.Sc. and M.Sc. degrees in electrical engineering from Qatar University, Doha, Qatar, in 2014 and 2016, respectively, and the Ph.D. degree in electronic and electrical engineering from the University of Strathclyde, Glasgow, U.K., in 2023. He joined Iberdrola in 2017, where he currently leads the power systems group at Iberdrola Innovation Middle East in Doha, Qatar. He has authored or coauthored around 30

articles in technical journals and international conferences, and received the Best Paper Award in the 2019 IEEE SGRE Conference. His research interests include power converters control, power system studies, power hardware-in-the-loop (PHIL) validation, and the techno-economic optimization of renewables and storage integration.



AHMED M. MASSOUD (SM'11) received the B.Sc. (First-Class Hons.) and M.Sc. degrees in electrical engineering from Alexandria University, Egypt, in 1997 and 2000, respectively, and the Ph.D. degree in electrical engineering from Heriot-Watt University, Edinburgh, U.K., in 2004. He is currently the Associate Dean for Research and Graduate Studies at the College of Engineering, Qatar University, Qatar, and a Professor at the Department of Electrical

Engineering, College of Engineering, Qatar University. He holds twelve U.S. patents. He published over 130 journal articles on power electronics, energy conversion, and power quality. Prof. Massoud's research interests include power electronics, energy conversion, renewable energy, and power quality. He supervised several M.Sc. and Ph.D. students at Qatar University. He has been awarded several research grants addressing research areas such as energy storage systems, renewable energy sources, HVDC systems, electric vehicles, pulsed power applications, and power electronics for aerospace applications



S. M. MUYEEN (S'03–M'08–SM'12–F'24) is a full professor in the Electrical Engineering Department of Qatar University. He received his B.Sc. Eng. Degree from Rajshahi University of Engineering and Technology (RUET), Bangladesh, formerly known as Rajshahi Institute of Technology, in 2000 and M. Eng. and Ph.D. Degrees from Kitami Institute of Technology, Japan, in 2005 and 2008, respectively, all in Electrical and Electronic Engineering. His

research interests are power system stability and control, electrical machine, FACTS, energy storage system (ESS), Renewable Energy, and HVDC system. He has been a Keynote Speaker and an Invited Speaker at many international conferences, workshops, and universities. He has published more than 450+ articles in different journals and international conferences. He has published seven books as an author or editor. He is serving as Editor/Associate Editor for many prestigious Journals from IEEE, IET, and other publishers, including IEEE Transactions on Energy Conversion, IEEE Power Engineering Letters, IET Renewable Power Generation and IET Generation, Transmission & Distribution, etc. Dr. Mueeen is a Fellow of IEEE, Chartered Professional Engineers, Australia, and a Fellow of Engineers Australia.



HAITHAM ABU-RUB (Fellow, IEEE) received two Ph.D. degrees. He has worked at many universities in many countries, including Poland, Palestine, USA, Germany, and Qatar. Since 2006, he has been with Texas A&M University at Qatar (TAMUQ) for 18 years and recently has joined Hamad bin Khalifa University while having an adjunct position as TAMUQ. For five years, he has served as the Chair for the Electrical and Computer Engineering Program, TAMUQ. He has established and served as the Managing Director

for the Smart Grid Center for ten years. Abu-Rub has published more than 600 journals and conference papers, eight books, and eight book chapters. His main research interests include electric drives, power electronic converters, renewable energy, and smart grid. Abu-Rub was a recipient of many national and international awards and recognitions and awards. He is a recipient of the American Fulbright Scholarship, the German Alexander von Humboldt Fellowship among others. Abu-Rub a Co-Editor in Chief of the IEEE Transactions on Industrial Electronics.